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MODELING AND SIMULATION

“Simulating Forest Fire using Cellular Automata”

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1. Introduction

Modeling and Simulation is a technique used to represent real-world systems in a virtual environment to analyze their behavior without affecting actual operations or risking real-world destruction. It is widely applied in environmental management and disaster response to study how variables interact, predict outcomes, and improve preventative strategies. Simulation is especially important in decision-making because it allows testing different environmental scenarios without real-world risks or costs.

This project focuses on simulating the spread of a forest fire using a cellular automata model in NetLogo. The study uses simulation to examine how environmental variables—specifically tree density and wind mechanics—interact to spread fire. By evaluating these factors, this simulation helps identify the mathematical tipping points where local fires transition into widespread disasters.

2. Problem Description

In many forested regions, wildfires have become increasingly severe due to dense, overgrown vegetation. When a fire breaks out in a highly dense forest, it can quickly escalate beyond the control of emergency responders, leading to massive environmental and property damage. Land management often finds it difficult to decide how much of a forest needs to be thinned or cleared to prevent these chain reactions. Therefore, a simulation approach is required to analyze the "percolation threshold" (the exact tipping point of fire spread) before making costly and labor-intensive real-world changes to the environment.

3. Objectives

The objectives of this simulation study are:

- To model a forest fire spread system using grid-based discrete cellular automata.
- To identify the critical forest density percentage (percolation threshold) at which fire spread transitions from a localized event to map-wide destruction.
- To analyze how varying wind strengths and directions alter the probability of fire spread.
- To compare system performance under low-density (safe) and high-density (doomed) conditions.

4. System Description

The system consists of a grid representing a physical forest landscape.

- **Entities:** The primary entities are the Trees (acting as fuel) and the Fire.
- **Resources:** The simulation grid (patches) acts as the available land area, containing either dirt (empty space) or trees.
- **States (Queues):** Trees transition sequentially through specific states: Healthy (Green), Actively Burning (Red), and Consumed/Ash (Black).
- **Events:** The system is driven by initial edge ignition, dynamic fire spread (where embers jump to nearby trees based on wind and distance), and fire exhaustion (when a tree's burn timer reaches zero).

5. Model Design

The simulation model is designed using the following logical structures:

- **Setup Module:** Generates the spatial matrix based on the user's selected density percentage, spawning healthy trees and igniting the western edge to simulate a fire boundary.
- **Process (Spread) Module:** Represents the fire's progression. Burning patches evaluate surrounding healthy trees, using a custom probability function that factors in wind direction, wind strength, and the inverse square of the distance to determine if the fire spreads.
- **Automation Module:** Utilizes an auto-run feature to automatically simulate the environment repeatedly from 0% to 100% density, tracking the entity flow from healthy to ash, and dynamically plotting the final damage percentage.

6. Assumptions

To isolate the effects of density and wind on fire percolation, several system limitations and rules were applied:

- Topography is assumed to be perfectly flat, ignoring how hills or valleys might accelerate fire.
- Fuel is uniform; all generated trees are assumed to be of the exact same species and dryness level.
- Burn time follows a randomized distribution between 10 and 30 ticks to account for natural variances in fuel load.
- Weather factors such as humidity, rainfall, and temperature are excluded from the mathematical probability of ignition.
- The fire is only initially ignited on the left (western) edge of the map.

7. Simulation Setup

The simulation is run dynamically until the system reaches its stop condition: zero actively burning trees remaining on the map. To account for randomness in tree generation and fire spread, the auto-run feature performs 3 replications for every density level, sweeping automatically from 0% to 100% density. Performance measures collected include the live percentage of the forest burned, total trees consumed, and the final total damage percentage. These measures provide a reliable estimate of the percolation threshold and system behavior.

8. Results & Analysis

The simulation results demonstrate a clear, non-linear relationship between forest density, environmental factors, and fire spread. During low-density conditions (0% - 50%), the average fire damage is extremely low. The fire quickly runs out of fuel and dies out safely, indicating that the system can naturally contain the fire because the distance between trees is too great for embers to travel.

Conversely, under high-density conditions (65% - 100%), the system experiences rapid fire spread, resulting in nearly 100% destruction of the forest. The exponential increase in damage is strictly due to the higher availability and tight connectivity of fuel, creating an unbroken chain for the fire to travel across.

Table 1. Result and Analysis Table Comparison

Scenario	Forest Density %	Wind Strength	Wind Direction	Total Trees Planted	Final Damage %
Safe Forest	30%	0	None	3074	1.85%
Doomed Forest	80%	0	None	8145	97.11%
Weak Wind (East)	50%	10	East	5054	5.16%
Strong Wind (East)	50%	50	East	5054	96.89%
Wind: None	40%	25	None	4100	1.56%
Wind: East (Forward)	40%	25	East	4100	6.12%
Wind: West (Backward)	40%	25	West	4100	0.83%
Wind: North (Sideways)	40%	25	North	4100	2.41%
Wind: South (Sideways)	40%	25	South	4100	2.8%

As demonstrated in Table 1, the simulation yields vastly different outcomes based on the interplay between forest density and environmental variables. The results can be broken down into three distinct behavioral categories: baseline density limits, aggressive/suppressive wind dynamics, and lateral wind dynamics.

The first two scenarios establish the natural mathematical boundaries of the forest without weather interference (Wind Strength 0). In a low-density environment of 30%, the fire is easily contained, destroying only 1.85% of the 3,074 planted trees because the fire quickly runs out of connected fuel. Conversely, when the forest density reaches 80%, the tight connectivity of fuel creates an unavoidable chain reaction, destroying 97.11% of the 8,145 trees. This confirms the existence of a critical percolation threshold governing system survival.

The simulation proves that wind strength acts as a catalyst that can artificially bypass the natural percolation threshold. At a moderate 50% density, a Weak East Wind (Strength 10) only results in 5.16% damage, as the wind is not strong enough to push embers across the empty gaps between trees. However, when the Wind Strength is increased to 50 (Strong

East Wind), the embers are violently pushed forward, bridging those empty gaps and causing massive destruction (96.89%). This demonstrates that high wind speeds completely neutralize the safety of a moderately dense forest.

The final five scenarios isolate the impact of wind direction by locking the density at a sparse 40% and the wind strength at a moderate 25.

- **The Baseline:** With no wind direction applied, the fire naturally destroys 1.56% of the forest.
- **The Accelerant (East):** Because the fire ignites on the western boundary, an Eastward (forward) wind pushes the fire into unburned fuel, raising the destruction rate to a localized high of 6.12%.
- **The Suppressant (West):** Conversely, a Westward (backward) wind fights the natural forward progression of the fire, pushing embers back against the starting edge and suppressing the damage down to a mere 0.83%.
- **Lateral Spread (North/South):** North and South winds force the fire to spread sideways along the Y-axis. This slightly increases the localized burning compared to the baseline (2.41% and 2.8%, respectively) but lacks the forward momentum required to force the fire across the map.

Ultimately, the data confirms that forest density establishes the natural baseline for fire risk. However, weather variables act as ultimate modifiers: wind strength determines how far embers can jump across safe gaps, while wind direction dictates whether the weather will act as a catastrophic accelerant or a natural fire suppressant.

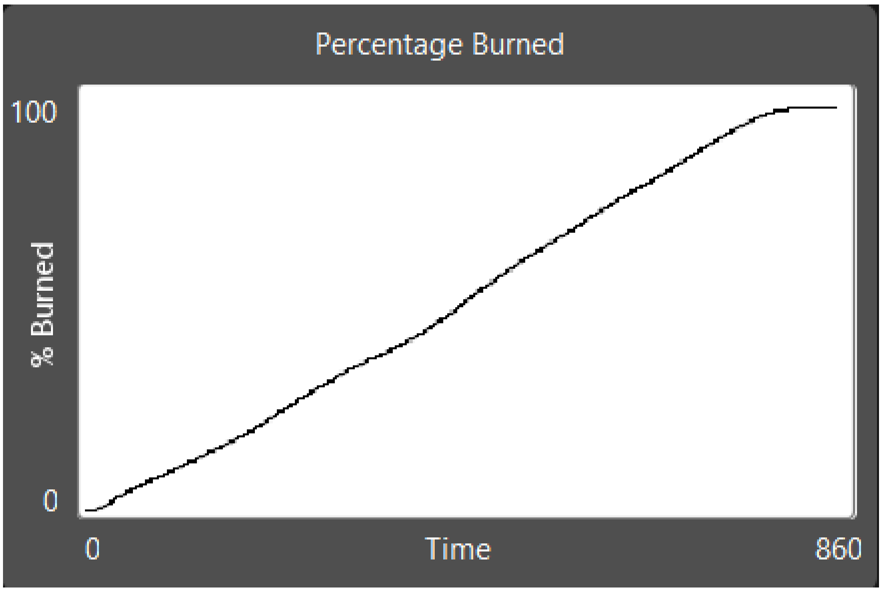

Figure 1. Comparison of Fire Spread Over Time

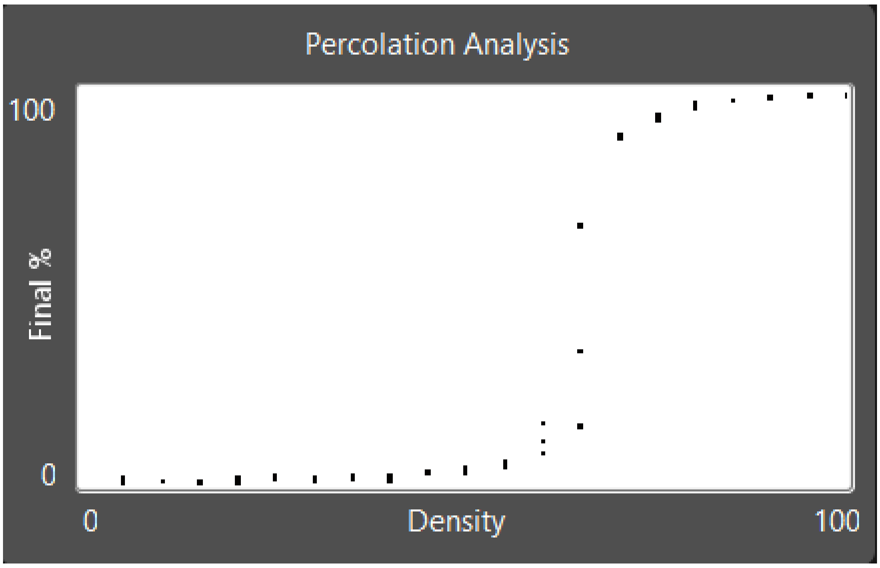

Figure 2. Queue/Percolation Tipping Point Comparison

The real-time data from Figure 1 highlights how rapidly a dense forest burns, with the "Percentage Burned" line climbing steeply until the entire system is consumed.

However, the most critical finding is shown in the Percolation Analysis graph (Figure 2). The data reveals a sudden, vertical jump in final damage (forming an S-Curve) occurring strictly between 55% and 60% density. This proves that fire destruction is not a gradual increase; it is a sudden chain reaction triggered the moment the environment hits this exact mathematical tipping point.

Furthermore, the simulation proves that weather conditions drastically alter this percolation threshold. As shown in Scenario 3, a forest at 50% density normally survives a fire outbreak with minimal damage. However, when a strong wind from the East is introduced, embers are pushed ahead of the main fire line. This allows the fire to jump across empty gaps it normally could not cross, completely bypassing the natural safety threshold and leading to massive destruction

9. Recommendations

Based on the cellular automata simulation results, the following strategic actions are recommended for forest management and emergency response teams:

- **Enforce Density Limits (The 50% Rule):** Park rangers and land managers must implement preventative measures, such as controlled burns and selective logging. Maintaining forest density strictly below the 50% percolation threshold in high-risk zones is the most effective way to prevent a localized ignition from becoming a map-wide disaster.
- **Dynamic Wind Monitoring:** Because the simulation proves that wind fluctuations significantly alter system performance and lower the natural safety threshold, continuous weather monitoring is crucial. Officials should dynamically escalate fire

threat levels during high-wind events, as wind allows embers to jump across safe gaps.
- **Strategic Resource Placement:** By utilizing weather data to anticipate how wind will push fire fronts, emergency responders can make better operational decisions regarding civilian evacuations and the placement of firefighting resources.

10. Conclusion

The simulation study successfully modeled the forest fire spread process using cellular automata and evaluated system performance under various density and wind conditions. The results show that fuel density has a significant impact on system behavior. During high-density scenarios, the system experiences exponential fire spread and total destruction, indicating that a dense forest struggles to naturally contain fire outbreaks.

Although more trees exist in high-density forests, the lack of empty space creates a continuous fuel chain. In contrast, low-density conditions demonstrate safer system performance, with localized fires dying out quickly. This shows that the environment is far more resilient under lower-density conditions.

Overall, the study concludes that the main challenge lies in managing the mathematical "tipping point" of the forest. While a forest survives adequately below 50% density, it becomes doomed to burn entirely when density increases past 60%. Therefore, land management improvements should focus on reducing fuel connectivity and maintaining density below the percolation threshold to achieve a safer, more resilient ecosystem.